Re:Wall

In what ways can the use of reclaimed materials redefine our approach to architectural design?

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ABSTRACT

The central theme of this report is to investigate how reused materials can impact architectural design. As we are in the midst of a global climate crisis and the building sector accounts for one-third of global emissions, redefining our building practice is paramount. Reused materials are one strategy for lessening the environmental impact, but they have not yet been incorporated into mainstream building practice. This project sought to show how reused materials can be utilised in a relevant and important way through architectural design adapted to reuse together with built prototypes in scale 1:1. To provide a foundation for the project, theoretical knowledge has been gathered through literature, case studies and conversations, while the speculative design work oscillated between 2D drawings, 3D models and physical explorations. The project was contextualised in Robertsfors in the north of Sweden, as the municipality is facing radical change due to the future construction of Norrbotniabanan and expects a housing boom in relation to that.

Some key findings were that reuse is of interest to actors all across the building sector and architectural field in northern Sweden, even when monetary gain is not part of the equation. The infrastructure for incorporating reuse is not established yet, but the municipality of Umeå has started an initiative called *Byggåterbruket,* trying to change that. Another significant lesson was that working with reuse is a very social endeavour, as a multitude of people had to be contacted in order to find reused materials, discuss collaborations or gain knowledge. Finally, reused materials have their own characteristics and limitations, and new strategies had to be developed both for the building aspect and for the design. The finished modular walls aim to show that it is possible and inspire others to explore reuse in building.

KEYWORDS

Reuse, Modular walls, Urban mining, Robertsfors

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GLOSSARY

CC – Centrum to centrum, a measurement common in building practice.

CO2e – Carbon Dioxide equivalent emission.

GDP – Gross Domestic Product, a key indicator of a country's economic health and activity.

GHG – Greenhouse Gases.

LCA – LCA stands for Life Cycle Assessment and is a common tool for measuring the environmental impact of a building and its emissions. Information and data are gathered from all stages of a building's lifespan, from material production to demolition and then compiled, giving an estimate of the building's emissions.

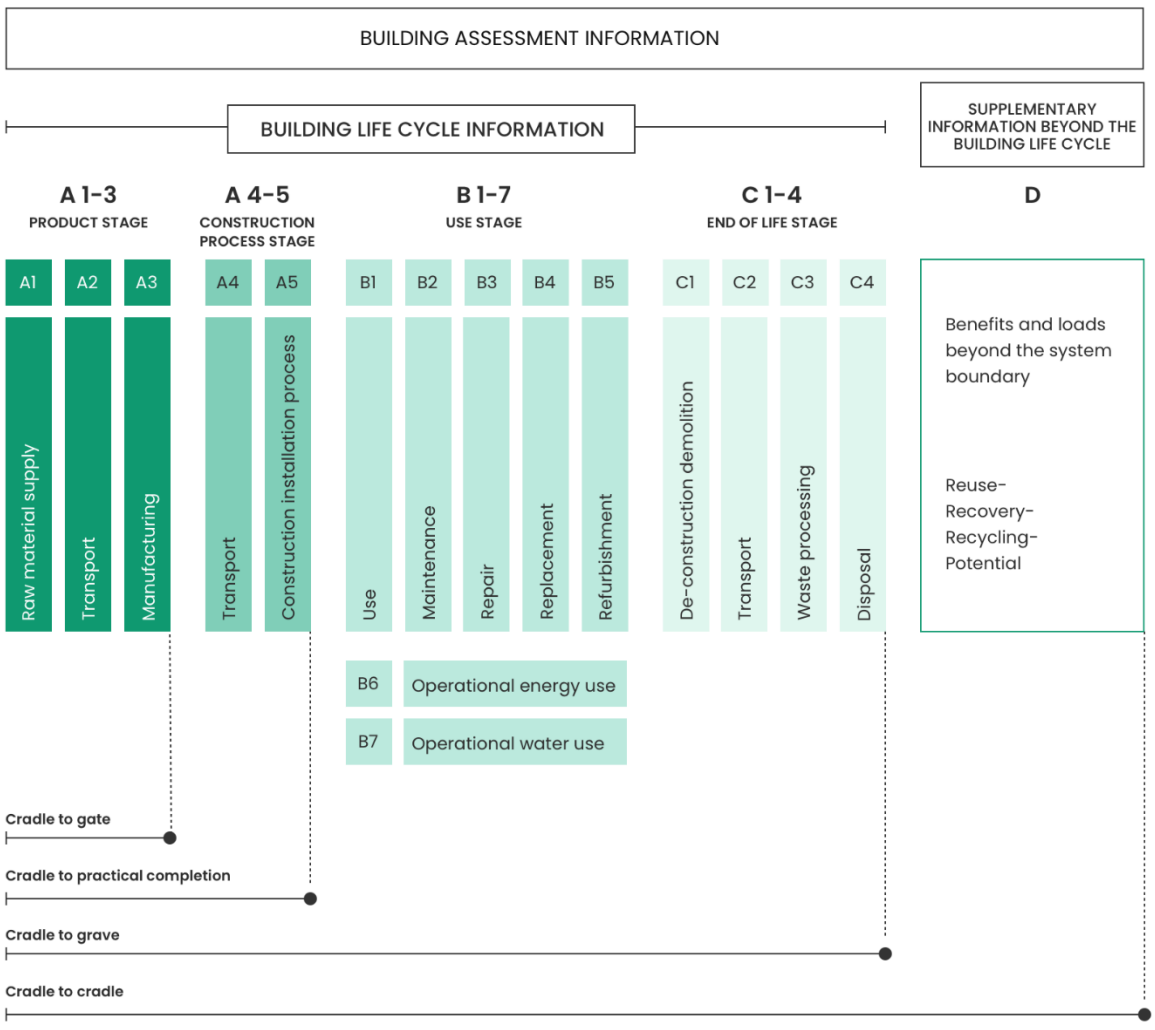


Figure 1. One Click LCA

The different life stages are categorised as:

A 1-3, Production stage

A 4-5, Construction process stage

B 1-7, Use stage

C 1-4, End of life stage

D, Supplementary information beyond the building life cycle

Embedded carbon – Embedded carbon refers to the CO2e released during stage A in the LCA, such as during material sourcing, transport or construction.

Operational carbon – Operational carbon refers to the CO2e emitted from stage B in the LCA, such as heating, cooling or electricity.

Reclaim – Taking a material, structure or item deemed as waste and repurposing it, restoring value to it.

Recycle – Taking a material, structure or item from its current context, manipulating and repurposing it into a new one, losing some of the structural integrity and inherent value.

Retrofit – Increasing the environmental performance of a building by adding new systems to it.

Reuse – Taking a material, structure or item from its current context and inserting it into a new one, retaining the structural integrity and inherent value.

Småhus – A term translateable to *small house*, a bit of an umbrella term for different types of housing such as detached villa, row house, chain house etc. Very unspecific and confusing definition to be found on Boverket and Skatteverket.

Urban mining – A practice where resources are not excavated from natural sources, but rather from the Anthropocene.

VAT – Value Added Tax, equivalent to the Swedish MOMS.

INTRODUCTION

This thesis began with an obsession of reclaimed materials. Where they are, how they act and how to reuse them. This topic then grew into issues of practicalities, legislative barriers and climate crisis, before culminating in a modular strategy for incorporating more reused materials into the building sector.

As the research continued, it became apparent that the building sector accounts for a significant part of the global GHG emissions, resource depletion and waste. Material reuse has entered the general debate in recent years as a necessary strategy for lessening the environmental impact of buildings, as well as increasing the lifespan of already existing ones. However, it is still evident that the practice of reuse predominantly occurs in urban settings, where projects benefit from substantial budgets or are driven by strong ideological motivations. Against this backdrop, the issue of reused materials was contextualised in Robertsfors, a small municipality with big dreams for the future. Through combining practical knowledge about the materials and their potential reuse with the genius loci, the possibilities of combining circular practices and architectural design were explored. Choosing Robertsfors was not only due to its geographical location or planned growth, but also the willingness of the municipality to collaborate and test new ideas.

The aim of this thesis is to facilitate better incorporation of reused materials in architectural practice. This was explored through the research question of *in what ways can the use of reclaimed materials redefine our approach to architectural design, resulting* in the development of a building system of modular walls tailored to the properties of reclaimed materials. By integrating circular building strategies and urban mining with a site-specific proposal, the project sought to provide knowledge, strategies, and insights on incorporating reuse within the construction industry in a concrete, practical, and contextual manner.

Divided into several chapters, this thesis zooms in from the global aspect of the climate crisis down to the very local contexts and 1:1 exploration. Something the project did not try to provide an answer for was LCA calculations, material quality control, or deeper explorations of anything other than the modular wall components. Case studies are introduced to illuminate the possibilities and struggles with reuse, and the potential for further research is included at the end.

DELIMITATIONS

When researching reused materials in the building sector, one quickly encounters LCA calculations. While this tool gives comparable and quantifiable answers to reuse and its implementation, it would be more fitting for someone with a Bachelor of Science in Building Technique to conduct the LCA-driven research.

It is also very easy to end up in the legislative part of reuse, rather than the design. There are so many obstacles for reuse, enough to make a whole thesis just about that. This project tried to actively refrain from getting caught in the legislative issues, instead focusing on the design-driven exploration so as to allow for the possibilities of reuse to be explored instead.

A similar issue is that of quality control and reassurances. To certify the modular walls built in combination with this thesis, knowledge far beyond that of an architecture student is necessary. Therefore, this project does not claim to create a finished product that is applicable immediately, but rather a product that challenges the status quo and all the barriers found in legislation and policies, letting people experience the possibilities of reuse in reality.

In order to reach sufficient depth for the design-driven research of this thesis, a conscious choice was made to focus solely on the building system for the facade walls, and not explore the flooring, roof structure or foundation in any significant way.

METHODOLOGY

To gain a better understanding of the topic, the research began with a review of literature on topics such as reuse, circular economies, urban mining, and architectural methods for reducing the environmental impact of the building sector. These sources led to further paths of investigation, as well as case studies relevant to reuse in architecture, which were analysed in order to identify strategies, possibilities, and barriers for working with reuse in practice. Simultaneously, a site analysis of Robertsfors was conducted through municipal planning documents, historical sources, site visits, and conversations with local actors, both municipal and governmental.

Speculative and research-based design work has been practised, where reused materials from demounted buildings and construction site surplus have been studied and experimented with to develop modular wall components. Since the project both explores theoretical strategies for incorporating reuse and formulates a finished design proposal, working with theory and practice in tandem was vital. The design process oscillated between theory, 2D drawings, 3D models and physical prototypes, and each iteration informed the next. By combining architectural investigation with material knowledge, the project has tested a possible future strategy for increased availability and approachability of reused materials within the building sector.

Most significant for this project methodology is the human connections that have been initiated. Conversations have been carried out with anyone from governmental employees to carpenters or demolition workers, and every piece of input has been valuable. The positive response the project has been met with was fantastic and made this whole process much more personal and filled with joy. Letting people from various sectors get attached and form a personal connection to a reuse project is, in my opinion, a very interesting strategy that could hopefully further the implementation of reuse in the building sector.

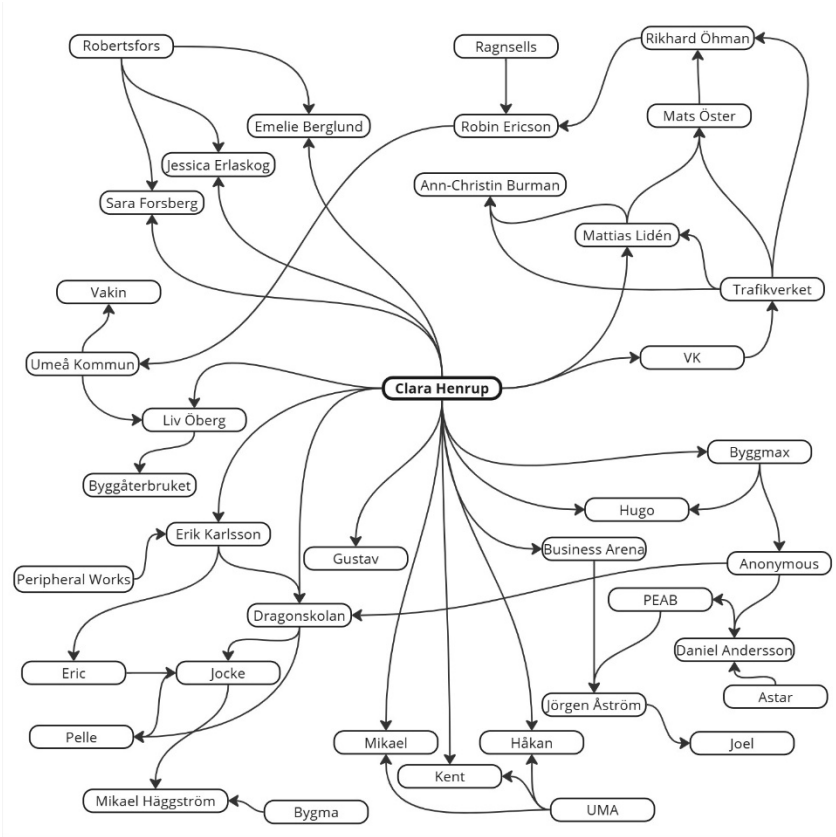


Figure 2. Network of human connections

CHAPTER 1 – CLIMATE CRISIS AND REUSE

In this chapter, the distinction between reuse and recycling will be clarified, the role of the building sector in the climate crisis will be emphasised, and the possibilities of a circular economy will be expanded on. To provide context to these topics, several sources from varying fields have been compiled, evaluated and summarised.

These sources collectively emphasise that today's linear “take, make and dispose” economy is completely unsustainable, especially within the building sector. *Cradle to Cradle*, authored by Michael Braungart and William McDonough in 2009, is a seminal work that has significantly influenced discussions on the Anthropocene impact on the environment, introducing circularity as not only an environmental strategy, but a core design principle to completely eliminate the concept of “waste”. In Duncan Baker-Brown’s *Re-Use atlas* from 2024, said design principles are implemented in real-life projects, with examples of reuse introduced at various stages of the architectural practice showcased through a wide range of case studies.

The United Nations Environment Programme *UNEP* serves as the premier international authority on environmental matters. *UNEP* regularly publishes technical reports, handbooks and guiding documents, two of which have been included in this chapter. The first report is called *Buildings and Climate Change: Summary for Decision-makers. It* gives relevant background on the environmental crisis and the role of the building sector, even though it is slightly outdated, as it was published in 2009. The second report is in collaboration with *Yale Center for Ecosystems + Architecture* and is called *Building Materials and the Climate: Constructing a New Future*. This report was published in 2023 and gives an updated insight into the building sector’s impact on the climate, and what we should do moving forward to mitigate the climate crisis. They add urgency and scientific backing to the issue, with a clear focus on the building sector and its impact on the environmental crisis, as well as highlighting the importance of material efficiency. *The Circular Economy and Benefits for Society* by Anders Wijkman and Kristian Skånberg for The Club of Rome is a study researching the economic, social and environmental impact on society when implementing a circular economy. The report shows us a prospective future through their circular economic model applied to several European countries. The outcome clearly showed that a shift to circularity decreased carbon emissions and increased job opportunities.

Significant across all these sources was that the distinction between reuse and recycling was important to make, as unconcise wording can be very detrimental in policymaking, laws and regulations, resulting in some of the challenges we face today. Recycling refers to the practice of taking something, be it an item, a structure or a material at the end of its projected lifespan and manipulating it to have a new purpose. The key item here is that the new purpose is of a lower grade than previously, and the item, structure or material has lost some of its original value. Reuse, however, entails taking said item, structure or material at the end of its lifespan, reconfiguring its purpose and creating a new life for it without lowering its inherent value. To exemplify this; imagine you have a piece of construction timber inside a structural wall that is at the end of its projected lifespan. If you choose to cut it up into smaller pieces and build a kitchen stool out of it, you have recycled it. If you instead choose to remove the timber from its current setting and insert it into a new wall, you have reused it.

To understand the urgency of reused materials in the building sector, one must understand the vast impact the buildings themselves have on the global climate crisis. In the report *Buildings and Climate Change: Summary for Decision-makers by UNEP,* it is estimated that buildings contribute to one-third of total greenhouse gas emissions. This is primarily through the use of fossil fuels during the operational stage of the building, although the production and transportation of the materials themselves make up for about 10-20% of the total emissions produced by a building. This view is further upheld and developed in the *Building Materials and the Climate: Constructing a New Future* by UNEP. The efforts of the building sector to reduce the amount of emissions produced during the operational phase are lauded, but the issue of forgetting to update the procedures for producing and transporting materials is highlighted. While the material part only

stands for 10-20% of the total emissions of a building now, once the operational phase becomes more efficient, that percentage will shift and become greater, thus requiring swift action.

After illuminating the environmental issues connected to the building sector and the need for circularity, it is also essential to look at the socio-economic consequences a circular economy brings. Wijkman and Skånberg's report, The Circular Economy and Benefits for Society, from 2017, is central for this, as it highlights the environmental gains of a circular shift and the possibilities for economic growth and job opportunities. In their simulation of a circular economy applied to several different European countries, Sweden included, all cases saw a significant reduction in CO2e emissions and increased job opportunities, resource availability, product lifespan and GDP. The authors estimate that if all aspects of their model were applied, 100.000 new jobs would be created, cutting the current nationwide unemployment by one third.

CHAPTER 2 – BARRIERS TO CIRCULAR BUILDING

Considering the dire state of the climate crisis, the urgent change that needs to happen and the building sector's part in it, how come reuse is not happening more often and at a larger scale? Some of the barriers I identified through my research entailed building codes and regulations not adapted for reused materials, economic disincentives, fragmented supply chains and knowledge gaps. One aspect I thought I would find as well was resistance within the building sector, but I was gladly surprised when that was not the case. The foreword in Baker-Brown's *Re-Use Atlas*, called Anything but Retrofit... by Prof Graeme Brooker, highlights the political and legislative blunders when dealing with reuse in the UK. The official government document *VAT notice 708* brings further understanding of legislation and policies in the UK, while the Swedish tax agency Skatteverket and the Swedish National Board of Housing, Building and Planning, Boverket, bring a local context to the issue. Personal communication from people within the building sector has been added to contextualise the problems mentioned, together with learnings from the municipal initiative *Byggåterbruket*, a circular marketplace for building materials.

Construction work incurs significant expenses in modern society. The costs of time, expertise and materials must all be considered. Contractors must strategically manage these elements to complete a project successfully while ensuring profitability, and any opportunities for cost savings or write-offs benefit the client or company financing the project. To steer the building sector and housing market towards the goals of the government, tax rates can be set accordingly. Embarking from the standard rate of 20% VAT, the government might need more housing or better energy performance, therefore regulating a 0% tax for converting houses to residential, or a 5% tax for installing energy-saving measures (Gov.uk, 2024).

As previously mentioned, having a clear and concise language when drafting these policies and legislations is paramount, as Prof Brooker explains that the word retrofit is regularly used interchangeably with reuse in UK legislation and policymaking (Baker-Brown, p. VII, 2024). This creates an issue as the complexities of working with reuse cannot be covered by the term retrofit, something that might explain the odd tax reliefs mentioned earlier. Using economic incentives to reach goals on a systemic level is obviously a successful strategy, but it can only be as successful and innovative as the goals themselves. This is why it is quite disheartening from a reuse point of view to see that in order to qualify for a 0% tax, pre-existing buildings need to be completely demolished, and a new building takes their place. If you reuse too much of the previous structure, the building is not eligible for the 0% tax addition (Gov.uk, 2024).

Sweden has a similar structure, although instead of tax rates differing, you can apply for tax write-offs in retrospect. The issue remains, though, that if you make significant changes, you can get the complete write-off, but if you make minimal changes, you get a partial write-off. Effectively, this means that you are rewarded for building a new house, repositioning piping and ducts or changing the floorplan, all of them labour- cost- and resource intensive procedures. But if you make smaller interventions, such as replacing the kitchen without changing the layout or replacing a material with something of equal or better value, like a tarp roof being exchanged for tiles, you are not rewarded to the same extent (Skatteverket, N.D.). Of course, minor changes fundamentally cost less than large interventions, but the intention of the tax write-off still sends a clear message of how reuse is valued within the government and the building sector. There are other economic incentives used in Sweden as well, such as environmental taxes, import-, waste- and natural resource taxes, together with VAT or subsidies. Similar to all of these approaches is that in order to make a significant impact, the tax level needs to be quite high from the beginning in order to economically promote circularity and reuse. Furthermore, implementing tax reductions and subsidies can have unforeseen results, as the economy is a very complex system (Boverket, p.70, 2024a).

The Swedish National Board of Housing, Building and Planning, also known as *Boverket*, is the governmental authority focused on building laws. To integrate reuse in a systemic manner, changes must occur at this level. Boverket identifies various building materials and structures currently used in the building sector, highlighting existing issues with reuse as well as potential future solutions. Common barriers include an immature market, a lack of knowledge, and fragmented material flows. Ensuring quality control for reused materials was a

challenge identified in almost all cases. While some materials, like concrete, have a finite lifespan due to carbonisation, others, such as wood and steel, can either maintain their quality or be treated to enhance their structural integrity. Boverket emphasised that quality control and reassurances regarding remaining lifespan should be assessed on a case-by-case basis (Boverket, 2024b). This method is more labour-intensive than using new materials, which is one of the many obstacles to reuse in the building sector.

Another issue that arises is the different legislation applied if the material is viewed as a product or waste. The Swedish Environmental Protection Agency, Naturvårdsverket, together with the Swedish Chemicals Agency, Kemikalieinspektionen, are responsible for the different regulations, and the overlap between them must be accurately controlled to ensure human and environmental safety (Naturvårdsverket, 2024). As reuse is essentially trying to take materials that would otherwise be considered waste and retain their status as product, or transform them from already classified as waste back to product, these legislations play a big role in material handling. Compare this legislative and administrative swamp of waste, quality reassurance and tax reductions with the straightforwardness of virgin materials, and it is quite understandable that circularity in the building sector is lacking.



Figure 3. Byggåterbruket

Umeå Kommun's initiative *Byggåterbruket* aims to address most of these issues brought forward in order to integrate reuse and circularity in the building sector. *Byggåterbruket* is a circular hub owned by an economic union called the *ABC group*, with members from multiple different sectors within construction. Here, companies can resell their surplus material, ensuring that materials fit for reuse do not enter the waste stream. The initiative offers storage, reselling and quality control of the materials they receive, quelling most of the obstacles previously mentioned. When approaching *Byggåterbruket* regarding what the benefits were for the companies in the *ABC group*, I was expecting money to be the driving factor. But the response I got was that the primary concern was reuse and facilitating a circular building sector in Umeå. Of course, there is a monetary gain to avoid landfill fees that would ordinarily be applied for throwing these materials away. Furthermore, when disclosing their environmental initiatives in their building process, being able to show that a certain percentage of material went to reuse can surely be beneficial for receiving grants, subsidies and tax reductions (Johansson, 2025).

Since *Byggåterbruket* has only been open since the middle of January 2025, there are still a lot of obstacles for this initiative (Johansson, 2025). Supply and demand have not yet reached a balance, and while there is storage space for materials sheltered from the elements, sometimes that is not enough for the volume of materials received, and sensitive material is stored outside, resulting in damage. Speaking to some carpenters, I got the feeling that the public image of *Byggåterbruket* has not yet reached a high enough standing to attract people in the business, specifically in regard to the storage difficulties (Holmström, 2025). Nevertheless, I believe that this initiative is the way forward and will become an integral part of the circular market of building materials in northern Sweden once it has been operational for longer.

CHAPTER 3 – CASE STUDIES

This chapter will introduce the case studies that have been significant for the project. *Svanen* and *Upcycle House* by Lendager Architects, together with *Hedlundaskolan* in Umeå and the works of *Rotor DC*. These case studies illustrate various methods of reuse and urban mining from a European viewpoint, together with the very local case situated in Umeå.

SVANEN



Figure 4. Northern facade view of the Swan

Svanen, or the Swan, is a kindergarten situated in Gladsaxe, Copenhagen, designed by Lendager Arkitekter in Denmark. It was built using a lot of materials from the school that was previously situated on the site, making it a prime example of urban mining. Before demolishing the existing building on site, the architects, together with the demolition company, made a mapping of the materials. The result of this approach was 100% reuse of roof tiles, 6000 tonnes of concrete reused and the tender process for selective demolition was improved, among many other impacts (Lendager, 2022).



Figure 5. Overhead view of the Swan

Not only were materials reused, like the roof tiles that became both roofing and facade, but the old schoolyard clock can be found in the orangery, and the dome from the observatory is now part of the playground. What makes this building extra special is that it is the first of its kind, e.g. utilising circular building processes, which achieves the Swan label. To achieve the Swan label, the product must meet certain thresholds, mainly related to non-toxicity and human-safe chemicals.



Figure 6. Elevation drawing

The project has been groundbreaking enough to warrant a report by the Danish environmental agency, where they analyse and compile the steps necessary in order to successfully build an environmentally friendly, circular building, which serves as a guide for actors within the building sector wanting to achieve similar results.

UPCYCLE HOUSE



Figure 7. View of Upcycle House

The Upcycle House is a single-family home made from a variety of secondary materials, such as upcycled corks, OSB boards and shipping containers. This project did not gather materials from the site in the same way the Swan did, but rather collected both building waste and domestic waste in order to create something new (Lendager, 2023).



Figure 8. Interior view of Upcycle House

The structural framework is made up of the two shipping containers, while the rest of the exterior is made up of recycled aluminium cans and facade panels made from post-consumer granulated paper. The interior is made up of recycled champagne corks that become the tiling in the kitchen, recycled glass becomes the bathroom tiles, and the walls are clad with OSB boards that are pressed together without the use of glue, an aspect that is very important when talking about recyclability.



Figure 9. Recycled champagne cork cladding



Figure 10. Cork cladding and OSB boards

The project was under a strict monetary budget and had a goal of achieving 65% CO_2 emission reduction, something the architects were hesitant about whether they would achieve or not. Due to the economic constraint, the building utilizes passive house techniques, alongside specific orientation and different climate zones in order to reduce utility costs.

Once the project was completed, the architects realized that their LCA calculations showed an 86% CO_2 emission reduction, with no additional economic cost. The architects themselves were surprised, both by the

result but also that more studios and builders do not utilize secondary sources, urban mining practices and upcycling initiatives while building.

HEDLUNDASKOLAN



Figure 11. Visualization of Hedlundaskolan by Arkinova

Hedlundaskolan is a municipal primary school located in central Umeå and housing around 140 students. However, the school is reaching its limits, which is why the municipality has decided to build an additional building as well as renovate and update the old one. To lessen the climate impact, as much material as possible is reused when constructing the new building or renovating the old one (Umeå Kommun, 2024). Reused bricks from the vicinity will make up the facade on the new building, while the floors of the old building will be re-varnished rather than remodelled. Additionally, the layout of the existing building will be kept as much as possible, in order to minimise moving of walls, pipes and ducts (Wennberg, 2023).

Furthermore, the structure for the new building will be completely made of wood rather than steel and concrete, further limiting the embedded carbon of the building. The schoolyard will see some improvements as well, as the students of Hedlundaskolan got to participate in the design, and two of their suggestions will be built: a parkour track and a zipline. Additionally, the municipality organised a building-reuse festival for the students in autumn 2024, where they got to design with reused materials, creating anything from fabric flags to sofas for the new school (Umeå Kommun, 2025).

Construction work began in 2024, and the new building and schoolyard are supposed to be inaugurated and put to use in winter 2025/2026. Once that happens, the renovation of the old building will commence, planned to be finished by winter 2026/2027 (Umeå Kommun, 2024). To see a municipally owned project include reuse to this extent is fascinating and can hopefully lead the way for future projects. However, I am slightly worried by the lack of information and PR reaching the general public. I have a genuine interest in reuse, and through active searching, I found only a few municipal articles of approximately 50 words each regarding Hedlundaskolan. To fully incorporate reuse in mainstream building practices, thoughts and examples of reuse must become a regular part of our collective thinking, not be a greenwashing quick fix that is then forgotten about. I think Hedlundaskolan is a step in the right direction; it is especially inspiring to see a Swedish case deal with reuse, and I hope we will continue to explore this topic fervently.

ROTOR DC

Rotor is a cooperative design practice based in Brussels, Belgium. To try and summarise this practice in one sentence is practically impossible as they have such a broad expertise, in anything from design, architecture and exhibitions, to demolition work, economic models and policy proposals (Rotor, 2025). Their focus and expertise lie in the reuse of building materials, knowledge they are actively seeking to expand with every

project they take on (Baker-Brown, 2024). In 2016, they initiated the spinoff project *Rotor DC*, which stands for *Deconstruction* and *Consulting*. Based on almost a decade of material flow research, this project trades in reused materials, carrying out a lot of the deconstruction themselves. Rotor DC offers facilities and consultation for processing, storing and reselling the materials, both online and in their physical shop.

Rotor DC's success can be attributed to their integrated approach, covering everything from deconstruction to resale, along with their multidisciplinary expertise. They offer storage facilities for reclaimed materials, which addresses one of the major challenges for reuse in the conventional building sector, providing reassurance to companies interested in reuse but hesitant to take risks.

I think *Byggåterbruket* in Umeå stands to gain significant insights from *Rotor DC*. As the project is currently in its early stages, a consistent workflow is still lacking. *Byggåterbruket* is highly dependent on its partners, leading to a fragmented inventory that is significantly impacted by supply, demand, and logistics. Once *Byggåterbruket* solidifies its presence in the market, enhances its expertise, improves logistics, and develops a reliable online platform, it has the potential to achieve similar success to that of *Rotor DC*.

These case studies paint a picture of possibilities in reuse. From the very contextualised examples from Lendager architects, where the projects have been successfully realised, to the systemic approaches of Rotor DC. Aforementioned cases are not from Sweden however, so including Hedlundaskolan brings the very local perspective to the table and illuminates how reuse is starting to establish itself in the global north.

CHAPTER 4 – ROBERTSFORS AS A LABORATORY

Robertsfors is a small municipality located between Umeå to the south and Skellefteå to the north. The municipality consists of many small villages as well as larger clusters such as Bygdeå, Sikeå and Robertsfors. The Norrbotniabanan railway is planned to go through central Robertsfors, creating a new meeting point at the railway station. Due to this massive infrastructure, the municipality is planning for population growth and more commuters, as central Robertsfors will have a swift connection to both Umeå and Skellefteå. To supply the relevant context for this chapter, several documents from Robertsfors municipality have been gathered, as well as newspaper articles and personal communication.



Figure 12. Map of central Robertsfors with the planned direction of Norrbotniabanan

Norrbotniabanan got the all-clear from the Swedish government on the 11th of February 2025 to commence the Dåva-Skellefteå part of the railway (Länsstyrelsen Västerbotten 2025). Consequently, certain villages and houses are in the direct path of the planned railway. The people affected by this were bought out by the Swedish Transport Administration, *Trafikverket*, and the buildings face demolition (Stormare, 2025). Utilising urban mining principles on these buildings seems perfect, as the structures obviously are in working condition, and the chance of the building materials being eligible for reuse rather than demolition seems high. When talking to actors within the demolition sector, it was made clear that most demolition work is

ordered due to damage from either moisture or smoke and therefore the materials are seldom fit for reuse (Ericson, 2025).

When I contacted *Trafikverket* regarding these buildings that were bought out and facing demolition, I got to take part in very interesting discussions and learn more about the significant role that policies and laws have in both demolition and construction. One house in particular had to be demolished before the 31st of March, as it previously had a colony of bats living in the attic many years ago (Öhman, 2025). Disruptive works, such as renovation, construction or demolition, were prohibited by the County Administrative Board *Länsstyrelsen* from being carried out during the summer when the bats would occupy the house. Since then, the attic had been rebuilt, and the bats had not lived there for years or even decades, but the prohibition remained. This meant that timing was of the essence for demolishing this building, and I saw an opportunity to join the demolition inventory carried out beforehand and identify materials with potential for reuse.

Regrettably, this process extended beyond the expected timeframe. As *Trafikverket* is a governmental agency, it must adhere to specific procedures when outsourcing work to prevent corruption and ensure equitable opportunities. While I was still invited to join the demolition intervention, the building will not be torn down in 2025 and subsequently will not be part of my design proposal, as the inventory has not yet taken place as of May 2025. However, Robertsfors is not the only place that will transform due to infrastructure, and I think this very scenario will happen again. With longer preparation and better timing, such a building could be perfect for replicating this project.

Choosing Robertsfors was mainly due to the future growth potential, together with a general air of excitement and “yes, let’s do something new!” that permeates the municipality. Introducing reuse seemed like a great opportunity in that social climate. An added bonus that has been found is the rich cultural heritage of the iron industry and garden city planning present in central Robertsfors, which has been a delight to design with and against, lending a form language to the design that allowed the project to be more focused on the materials themselves.

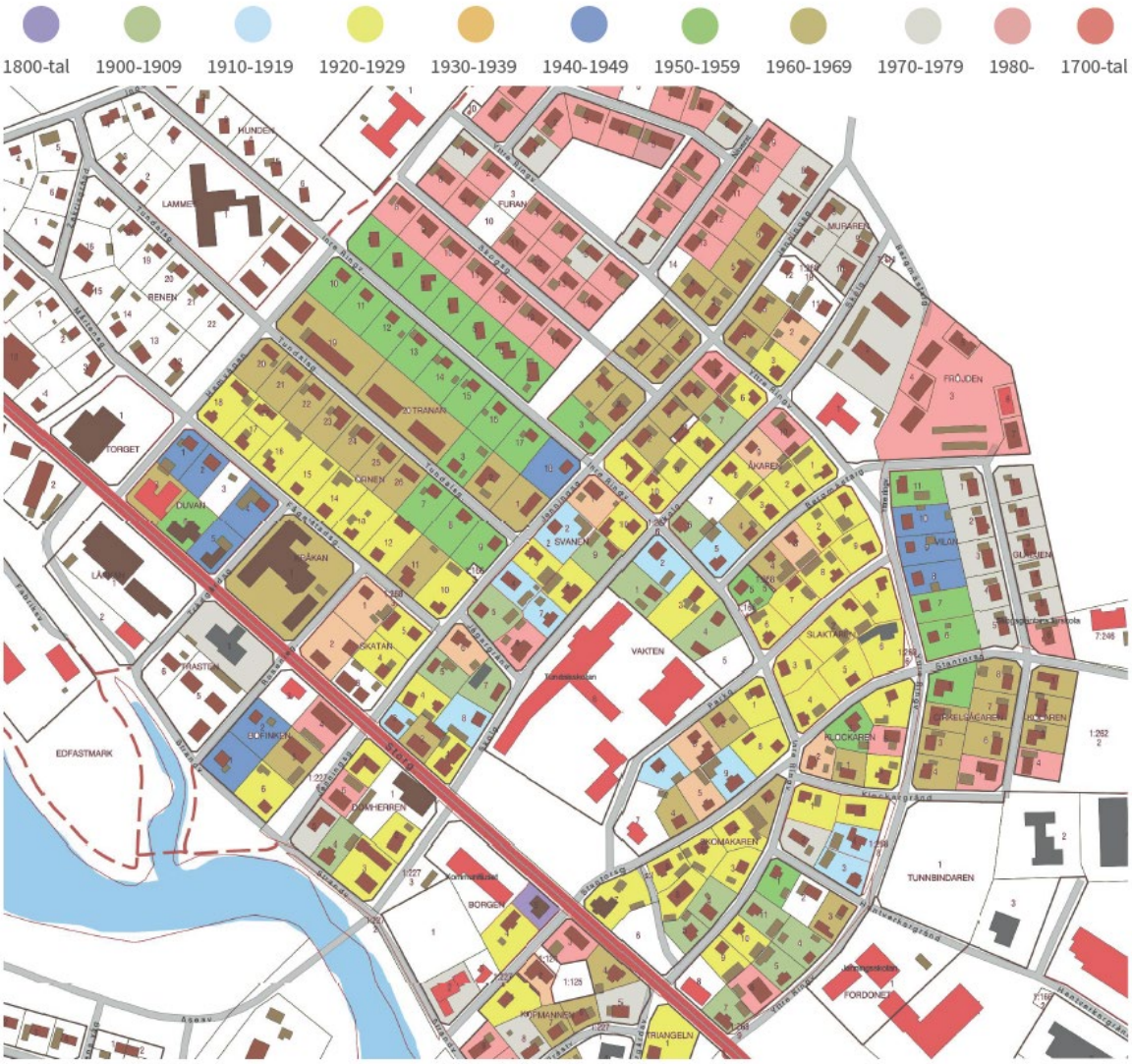


Figure 13. Map of småhus in central Robertsfors

Central Robertsfors is littered with old *småhus*, ranging from the 1700s to the 1980s, and most of the area is recognised as important cultural heritage under protection (Robertsfors Kommun, 2022). Walking through the winding streets, you are transported back in time, and it feels like a history book coming to life. The municipality actively wants to retain this characteristic, together with the garden city heritage, as they specified in their planning document, *Planprogram Robertsfors 2022.* The new areas should follow the same size of housing, as well as city layout and ratio between garden and house per plot, with the addition of some larger multi-family houses along the central roads to generate more movement (Robertsfors Kommun, 2022). This intention is further upheld by the detail plan for the area indicated in fig. 14, as it will accommodate larger houses taking up more space per plot (Robertsfors Kommun, 2020).



Figure 14. Map of central Robertsfors with my area of investigation

CHAPTER 5 – DESIGN PROPOSAL

From the beginning of this project, I knew that I wanted to work with reused materials, showing that they can be used in a relevant and important way. I also wanted to design something that was reversible, inspired by the 12 ISDA principles of Design for Environment (Baker-Brown, p.XIV, 2024), so for instance all the joinery is made with screws rather than nails. My design proposal became a *småhus* made from reclaimed materials, fitting into the context of Robertsfors. This entailed working both in the digital realm, developing a design proposal adapted to reuse, and in the physical realm, gathering reused materials and building two wall components on a 1:1 scale for the design. My hypothesis going into the design phase was that the digital design would be the driving factor, and that the physical test would just be a visualisation of sorts. I was unprepared for how much I would learn from building in scale 1:1, the difference you experience when going from digital to physical, and how vital the lessons learned would be for my structural strategy.

As my project is situated in Robertsfors, Västerbotten, and the materials I gathered were from the vicinity, wooden beams became my structural skeleton, and wooden facade planks the cladding. Working with physical prototypes gave me valuable insight into the characteristics and limitations of the materials that cannot be accurately represented in a digital model. For example, some of the beams I gathered had been stored outside, resulting in them getting bent, and it was difficult to find material of sufficient length and dimension. These limitations sparked the idea of modular wall components, circumventing the issue of length and instead enabling the application of reused materials. The issue then became how to join these modular walls, which was tested through conceptual models.

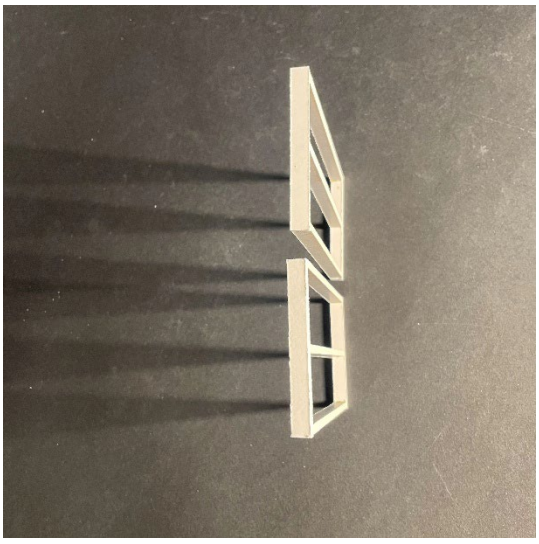


Figure 15. Model

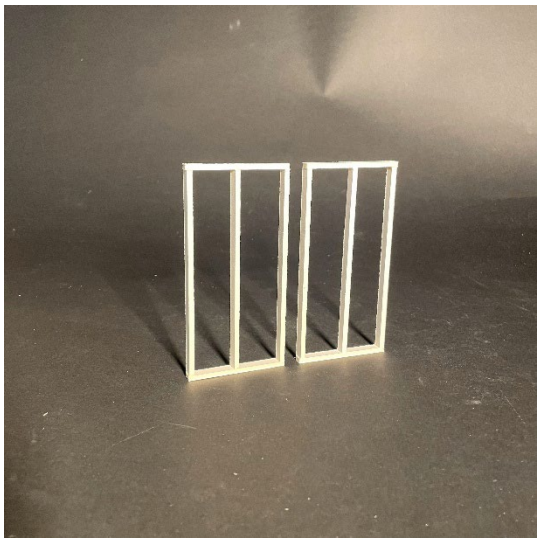


Figure 16. Model

Letting two elements just meet head-on was one possibility, and quite similar to how regular walls are built. This solution would require extra reinforcement of the horizontal beam; however, for stability, the joints would be at risk for creating thermal bridges, and I felt like this was too similar to standard building practice. I wanted my project to test, explore and question the use of reused materials, furthering our knowledge rather than doing business as usual.



Figure 17. Model

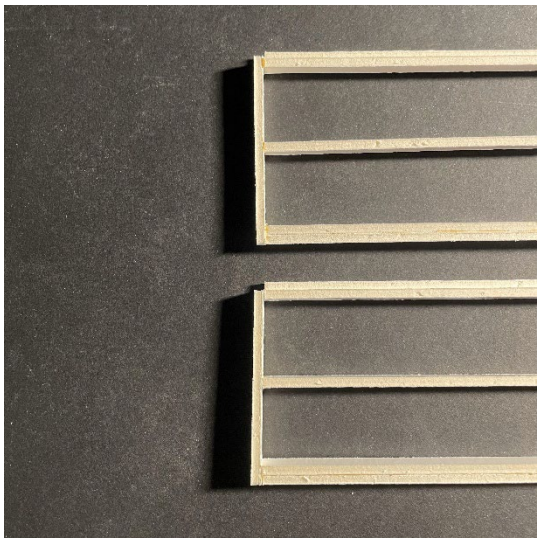


Figure 18. Model

So instead, an extra vertical beam was introduced, cut in half to create a puzzle-piece joint of sorts. The joint ensures that wind and cold temperature cannot enter the building as easily as the previous suggestion thanks to the overlapping beams, also lending structural stability to the walls without extra horizontal reinforcement. But while the puzzle-piece joint is quite elegant for walls standing in succession, the corners of the building quickly became a problem.



Figure 19. Corner solution

Here, working with physical prototypes proved vital again. To have the beams in 1:1 scale, together with some preliminary iterations out of scrap pieces, I could literally puzzle them together and test different solutions to the corners. Ultimately, this iteration felt the most relevant, where you take the same modular wall component but add an extra puzzle-beam on the inside of the wall. This ensures that the perpendicular wall has something to attach to, as well as the interior cladding boards. The puzzle-beam of the perpendicular wall is unscrewed, flipped 90 degrees and then reattached, ensuring a flush corner. These walls used beams of a 145 mm x 45 mm dimension and resulted in quite easy and elegant joinery, but one must keep in mind

that beams of thicker dimensions might not behave the same, resulting in additional manipulation of the materials being needed.

However, the challenges with reused beams did not stop at the dimensions and joinery solutions; they continued with the bent pieces I got. I measured, cut up, and placed the beams I needed for my two wall components, and it was recommended that the straightest pieces be located along the frame of the component. This would ensure the structure was as straight as possible while also giving me the possibility to force the middle beam straight. With the help of a metal pole, heavy duty screws and a clamp, together with some extra manpower, I was able to force the bent element straighter and lock it in place. Unfortunately, almost all of the beams were bent in one direction or another, so the finished structure was not straight anyway.



Figure 20. Skeletal walls from reused materials

This did not present a problem until I had added the insulation, wind- and vapour barrier and had arrived at the stage where interior cladding would be fastened. In a continuous wall, a certain angle could be okay as long as it is compensated for in the end, but since these modules are so small, issues become relatively large, and the space to compensate for them is very limited. Since the interior cladding was made from OSB boards

and had 90-degree angles, the skewed angle of the wall skeleton did not align. To solve this issue, I was introduced to the creative use of ratchet-straps, quite literally forcing the opposite corners closer together. Once a straight angle was achieved, the OSB boards were screwed to the structure, locking the wall in its straightened position.



Figure 21. Wall straightening strategy

The structure follows the standard measurements of building materials, such as insulation being adapted to a cc measurement of 600mm, and OSB and plaster boards aligning with a cc measurement of 1200mm. Together with the additional puzzle-beam introduced, each wall module had a width of 1335mm, and the joints were located along a 1290mm grid. This grid influenced my final design heavily, together with the general dimensions of houses in Robertsfors and the plan drawings from Tullstugan in Stockholm by Mats Egelius (Egelius, p.58, 2018). While the facade walls had to align with the 1290mm grid, the interior walls could follow the same logic or align with the middle beam located at cc 645mm, ensuring a certain flexibility.

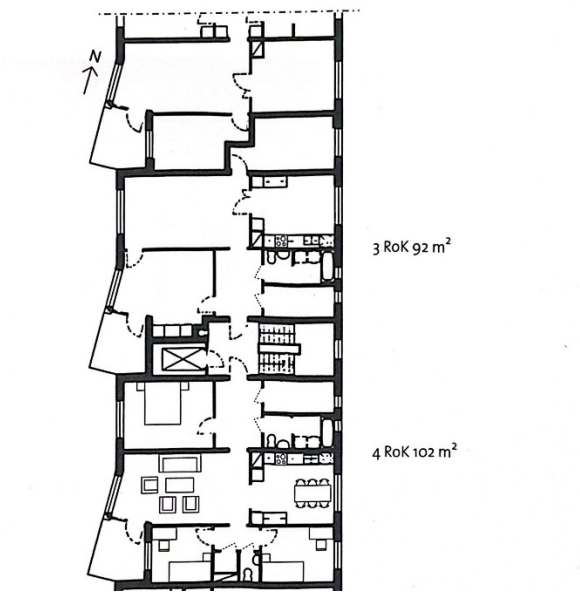


Figure 22. Tullstugan plan drawing

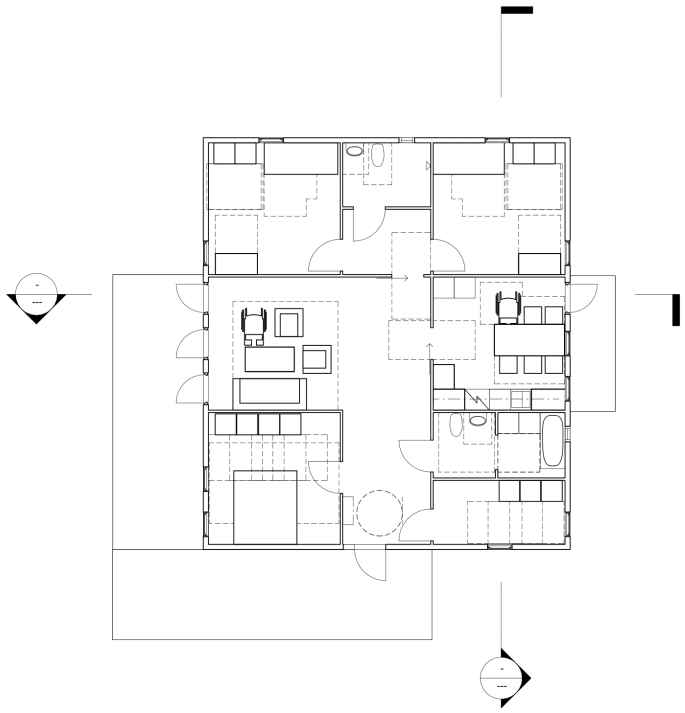


Figure 23. My design proposal, floor plan

The proposal has a somewhat squarer layout compared to the surrounding houses and the Tullstugan plan drawing, to account for the grid of the modular walls. You enter from the street facing south, stepping onto the continuous wooden deck before entering the hallway. The house is adapted to the highest standard of availability as prescribed by the Swedish Institute of Standards *SIS*, except for the secondary bathroom (SIS, 2006). One of the bedrooms greets you to the left, while a room with a flexible programme and the primary bathroom appear to your right. As you continue further into the house, the kitchen faces east and introduces a small outdoor space where you can sip your morning coffee, greeting the sun.

To the west, we find the living room, which opens up to the continuous wooden deck through multiple glazed doors. The reason for not introducing a continuous sliding patio door here is to stay true to the modularity of the project, only introducing elements that can fit into the wall modules. At the end of the house, there are two more bedrooms located, together with the secondary bathroom. This part, as well as the kitchen, has the possibility of being partitioned off by a sliding door. This is in response to the ideas of flexibility introduced by Anna Braide during a lecture, where the possibility to close off spaces or whole sections of a house allows for more programmes and activities to take place simultaneously (Braide, 2025). For example, trying to accommodate a sick preschooler, a parent cooking food, another parent working from home and joining a digital meeting with a child studying can be challenging in an open floor plan.

Although this floor plan indicates one master bedroom and two single bedrooms, all bedrooms are of equal size and can be adjusted to suit the requirements of the occupants. This flexibility is further visualised in the exterior cladding of the building, as the visual expression is finished by a wooden facade with horizontal planks and a cover strip panelling. Made out of reused planks demounted from an old barn in the vicinity of Skellefteå, these panels give the building a clear character as well as local contextualisation, while playing with the visual expression of different lengths and layouts.

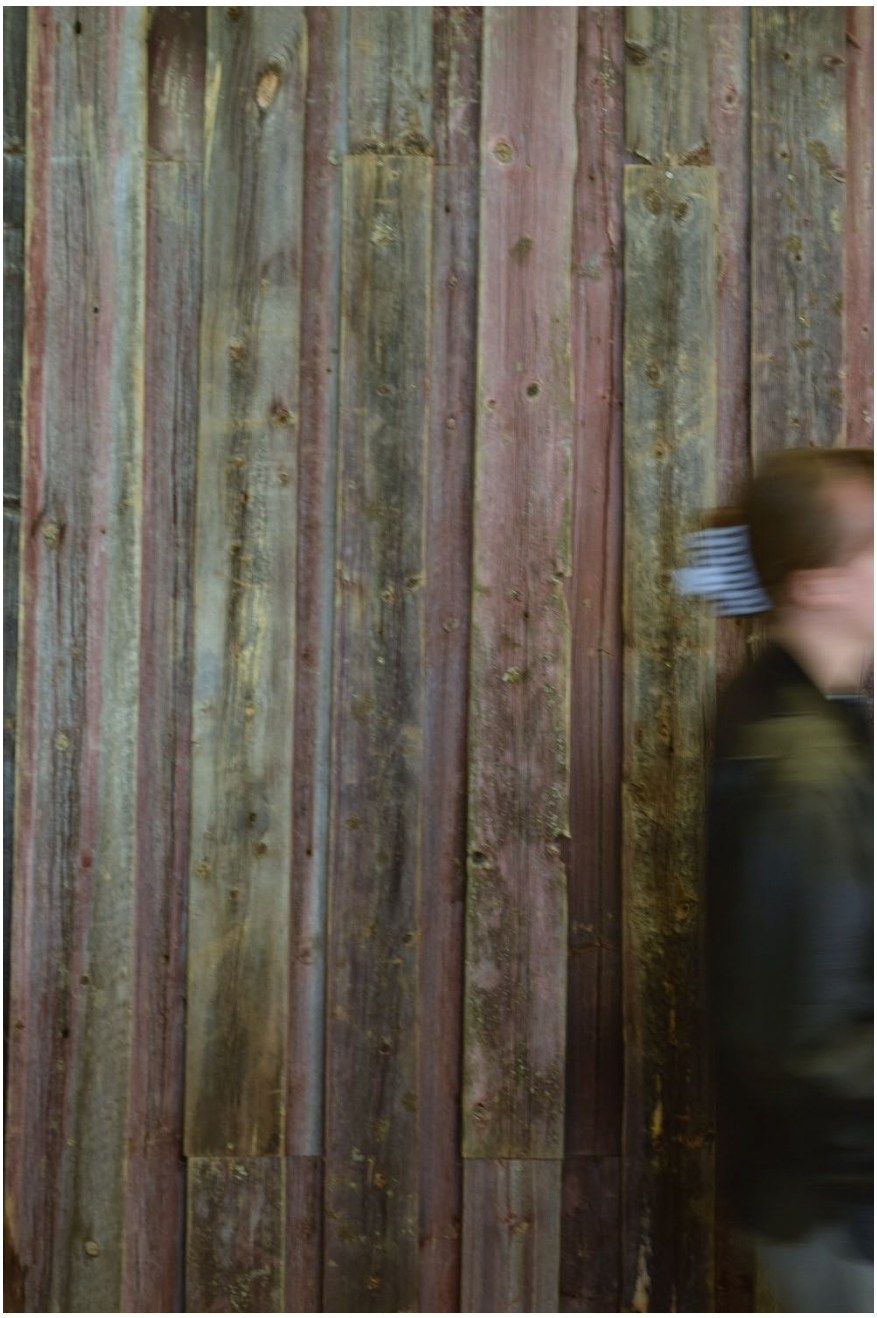


Figure 24. Wall with reused facade

While this project was adapted and influenced to fit the context of Robertsfors, the modularity of the walls entail that buildings can be scaled both up or down depending on the project. I do want to mention that this was more of an exploratory build rather than a certified building method. A counterargument to the limitations, regulatory issues and barriers currently found within the discussion on reuse. Further potential and research will be discussed in the next chapter, but I conclude this one with some lessons learned from my project.

Reuse is on the rise, entering the general debate. The facade planks, for instance, were not originally meant to be sold, but rather thrown away as the owner did not think there would be any value to speak of, but with the rise of reuse, they decided otherwise and sold them on Facebook Marketplace. There is a certain beauty to reused materials; they bring patina and history to the visual expression. If we are to incorporate reuse into the mainstream building practice, we must be open to exploring new strategies.

CHAPTER 6 – FURTHER RESEARCH

This thesis has examined how reused materials can be made available and approachable within small-scale and self-built construction, with a specific focus on geographical context and material-driven design. While the project finds that there is great potential in working with reuse in the global north, especially in municipalities facing industrial change, there are multiple paths to be explored further.

Primarily, adding Life Cycle Analysis to the process would be the most significant step forward. LCA calculations are a powerful tool to use for quantifying the environmental impact buildings and components have, and would give empirical weight to the project. By comparing and contrasting different material choices and construction systems, we can make visible the actual environmental impact they have and argue with scientific backing the urgency of incorporating reuse in the building sector. This is vital for the future potential of reuse projects, both in research and in the mainstream building sector.

While I had the ambition to incorporate LCA calculations in my project, I lacked the knowledge and resources for it. There is a whole course dedicated to it in Umeå University, as part of the Bachelor of science in building technique (Umeå University, N.D.), and while I think it is good that the issue gets significant time dedicated to it, I also mourn a little that it is not included more in the architectural practice. There are several online tools to simplify the process, though, such as One Click LCA (One Click LCA, 2025) and Speckle LCA (Link Arkitektur, 2025). The former is a widely used global tool, but it costs quite a bit of money, while the latter is an open-source web app developed by Link Architecture that is intended for early-stage design work. I think that reuse is the future, and architects will have a significant role in incorporating it into our built environment, which makes these tools very important for the future of our sector.

Another aspect worth researching more deeply is that of policy making and laws regarding reuse. Throughout the project, it has been clear that laws and regulations about demolition, permits and taxes impact the possibility and attractiveness of reuse within the building sector. Researching how these barriers can be solved, simplified or even circumvented would create better conditions for circular building practices.

Finally, there is a need for further design-driven research. How does the history of reused materials, their patina and irregularity, affect the architectural design process? The fragmented availability of materials and items, is that a hindrance or a possibility? How can new typologies and construction strategies be developed based on the characteristics of reuse, rather than imitating new-built projects? Through researching these areas, the architectural practice can become a significant actor in the transition towards a circular practice and sustainable building sector, not only in the major cities of Europe, but everywhere.

CONCLUSION

So, *in what ways can the use of reclaimed materials redefine our approach to architectural design*? My findings show that reused materials are not only a question of technical issues, but of spatiality and design as well, where the history and characteristics of the materials can be given new meaning through careful design. The global context of the climate crisis and the building sector's part in it was introduced as a starting point, and reuse was researched as a solution to that. Laws and regulations were presented that hinder the implementation of reuse, together with logistical barriers identified in northern Sweden. However, rather than seeing the limitations of reuse, we should embrace the challenge and find strategies that celebrate these characteristics, something this project aimed to do through the design of a modular wall construction system. To contextualise this proposal, Robertsfors was introduced as a testing ground as they face significant growth in connection with the new railway, Norrbotniabanan.

Several unforeseen lessons were learned along the way. One of them is the practical difficulties of working with reclaimed materials, as they do not conform to straight lines and perpendicular edges, together with creative solutions to negotiate these difficulties. Another is the absolute delight one can find within reused materials, how their history can shine through, and the human connections forged when sourcing these materials can bring joy, knowledge, and fascinating conversations. But most of all, the general air of excitement and interest I was met with when introducing my topic and aim to various people within the building sector, municipalities, and governmental agencies. My cynicism about people only doing things to make a profit was challenged as I experienced boundless generosity, and I am actually hopeful that the future of reuse in Västerbotten will be bright.

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